

Rheology of Basaltic Lava Flows from Pacaya with Stress-Strain Steps: A Laboratory Approach

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Introduction: Flank Collapse

- Stress-strain perturbations induced by magma intrusions and earthquakes can destabilize volcanic flanks, potentially trigger a collapse.
- At Pacaya, a flank collapse approximately 600-1500 years ago generated a 0.65 km³ debris avalanche that traveled ~25 km from the cone, also triggered eruptions due to depressurization in the magmatic system.
- Recent studies have identified instability on Pacaya's southwest flank, an area that has experienced the majority of eruptive activity since 1960, after a 200-year period of dormancy (**Fig. 1**).

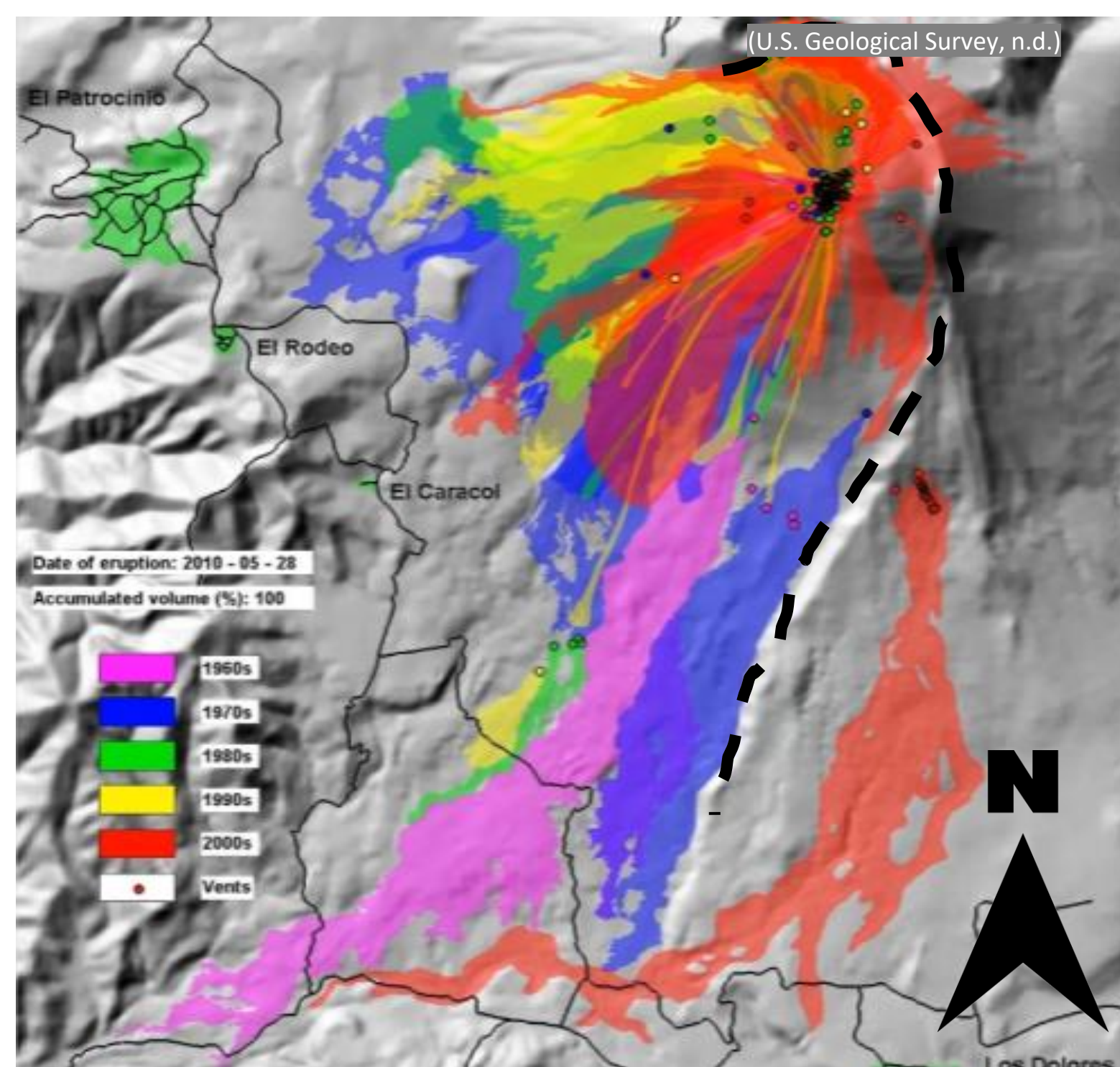


Figure 1: Lava flows map of the southwest flank of Pacaya, with the black dashed line indicating the previous collapse scarp.

Research Questions

- Do variations in physical properties influence the mechanical behavior of rocks?
- Can simple rheological models adequately capture a rock's mechanical response, or is additional complexity needed? If so, what complexities must be introduced to better match the data?

Importance

- While incorporating a heterogeneous internal structure in flank stability models brings them closer to real-world conditions, significant uncertainties persist (**Fig. 2**).
- Models commonly used fail to accurately capture the mechanical behavior of rocks by assuming constant mechanical properties through the deformation process.

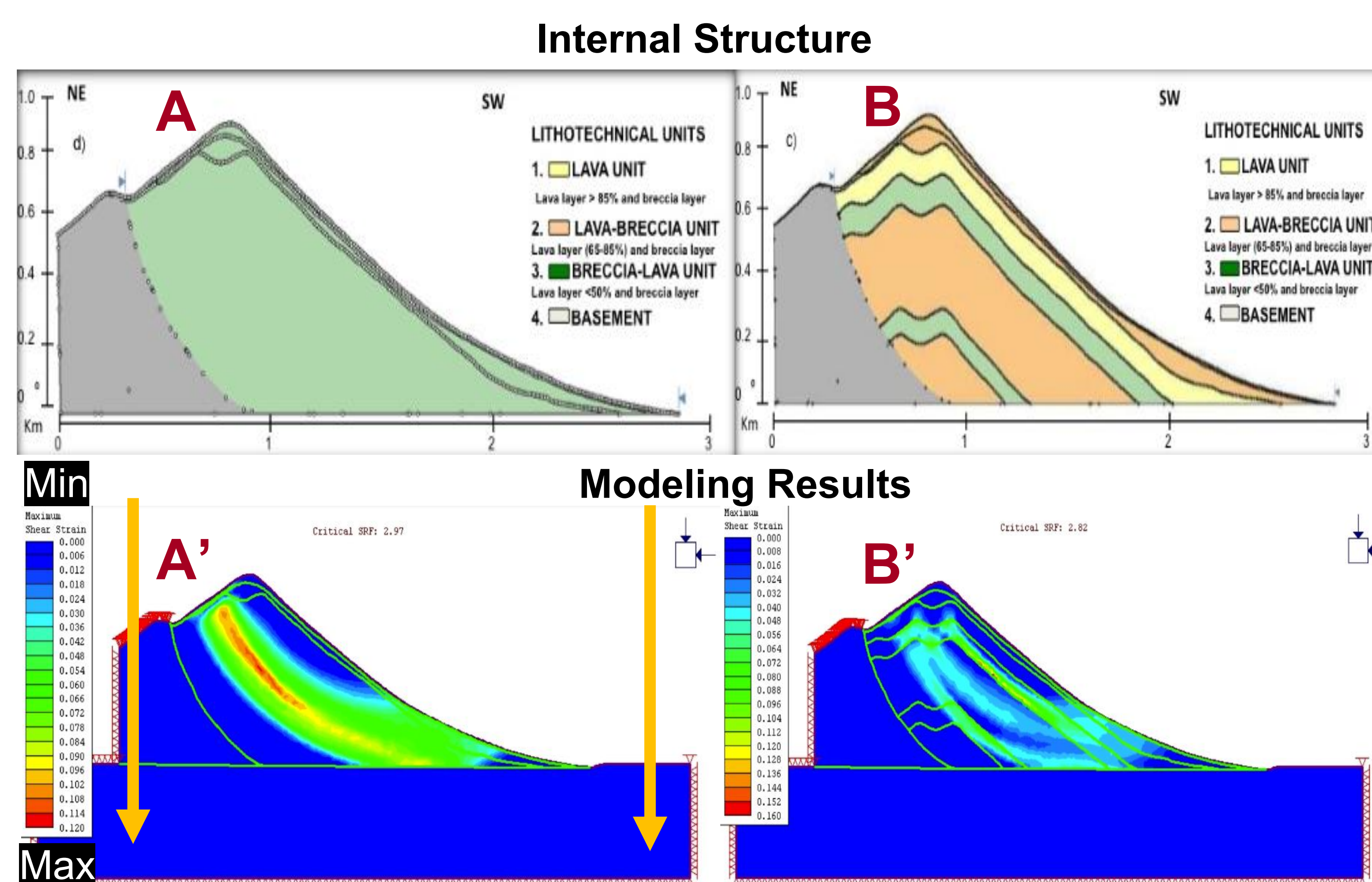


Figure 2: Flank stability models of the southwest flank of Pacaya volcano (Manzoni, 2012).

Rock Deformation Experiments

- Six deformation experiments were conducted using a uniaxial setup to collect strain, stress, and acoustic data.

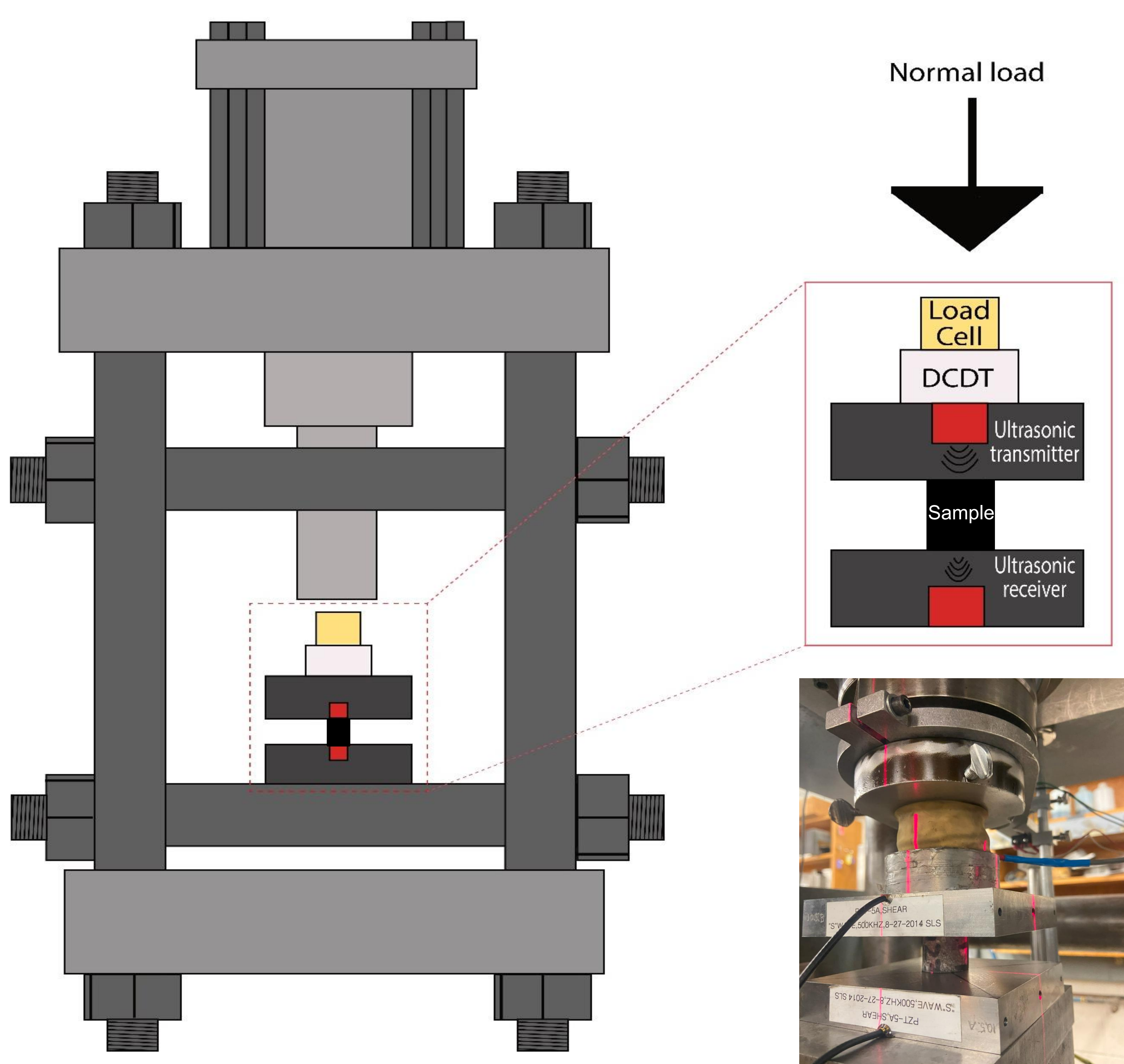


Figure 3: Uniaxial setup used for deformation experiments.

Experimental Results for Different Representative Samples

- The samples exhibited a broad range of physical properties, providing a comprehensive representation of different sections of a lava flow (**Fig. 4 & 5**).
- Variations in these physical properties are reflected in the results of the deformation experiments (**Fig. 6 & 7**). Different creep behaviors can be observed on each sample
- The remaining three experiments are additional representations of the samples presented.

Modeling Methods

- Three stress models were applied: linear elastic, non-linear elastic, and non-linear viscoelastic with changing viscosity (**Fig. 8**).
- The model parameters were estimated using emcee, a Python library for Markov Chain Monte Carlo (MCMC) sampling. To enhance the efficiency of the MCMC sampling, Optuna was used beforehand to optimize the initial parameter guesses through Bayesian optimization.

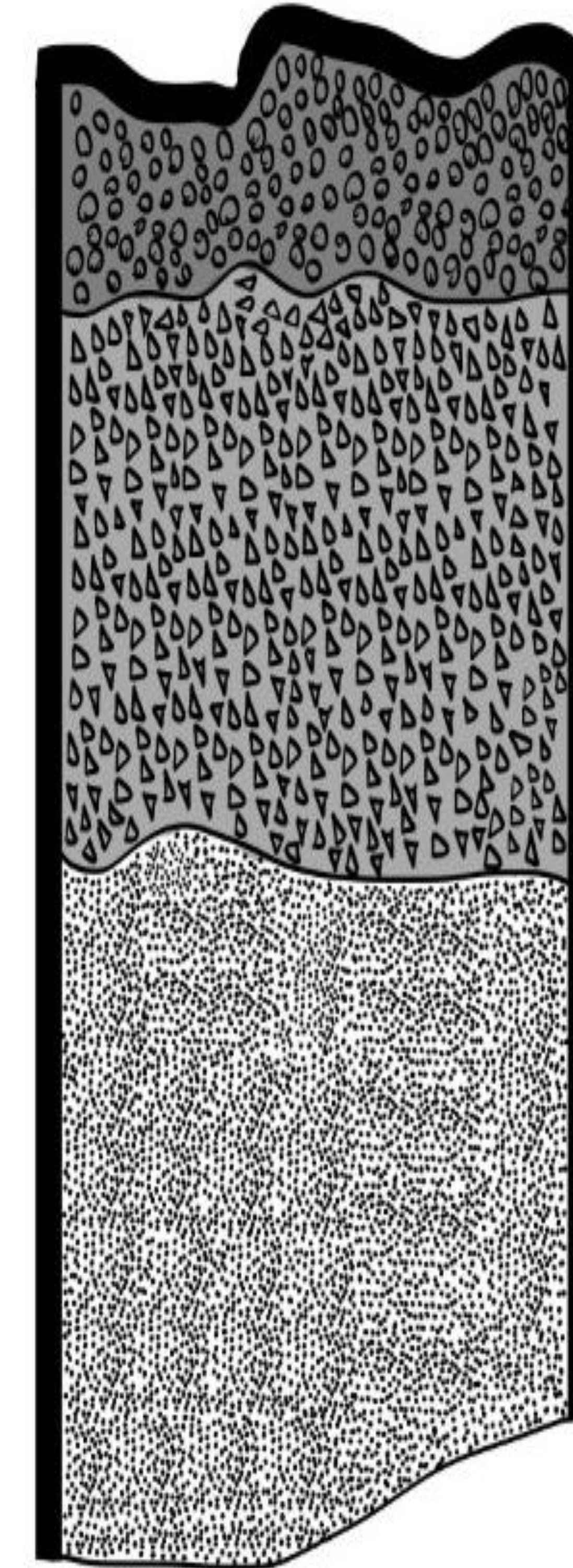


Figure 4: Schematic of lava flow core. **Figure 5:** Core samples.

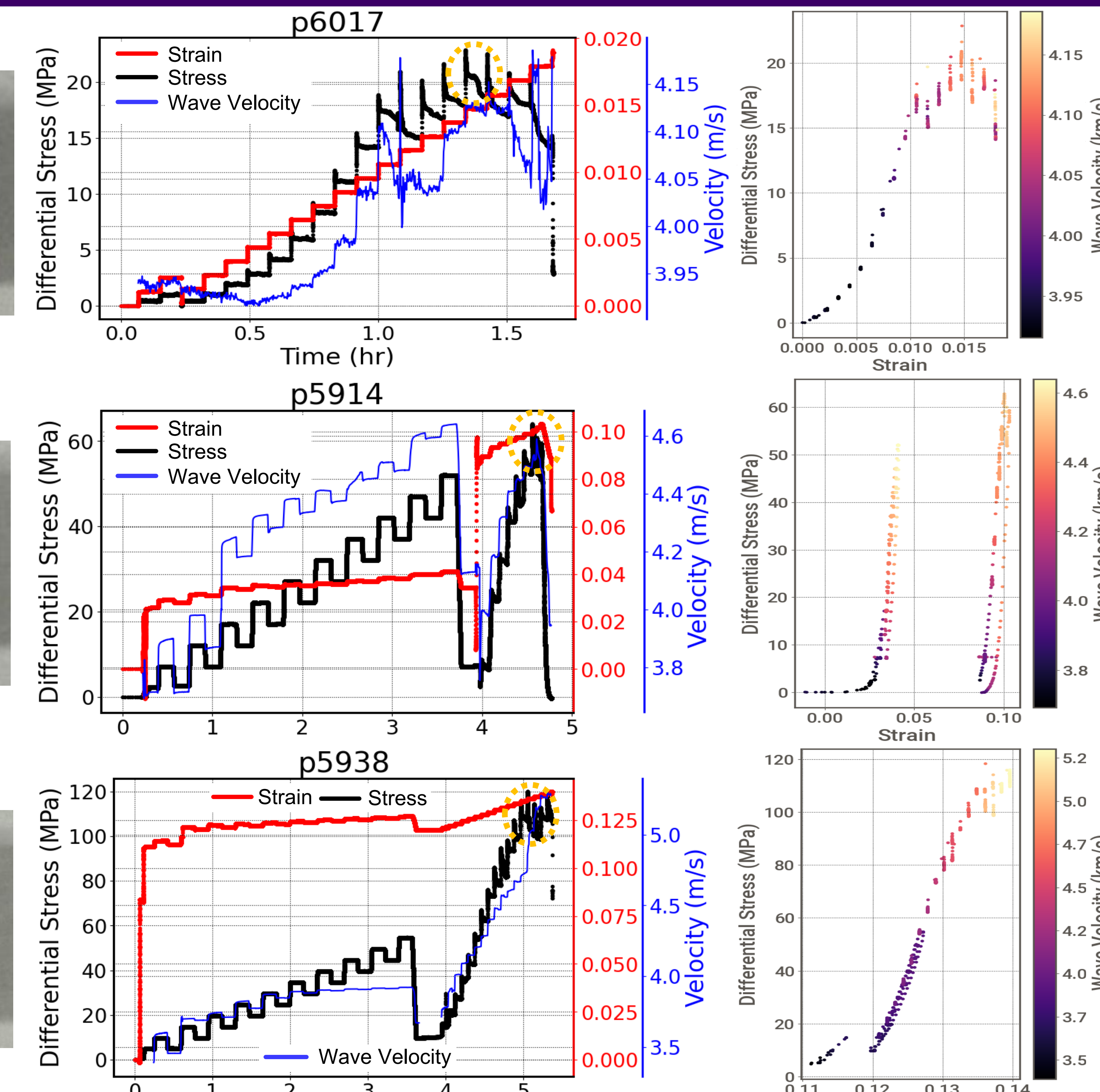


Figure 6: Mechanical and acoustic data, with the orange dashed circle showing the general area where tertiary creep begins. **Figure 7:** Stress-strain plot with wave velocity

Stress Model

Elastic

Viscoelastic

$$\sigma(\sigma, \varepsilon, v_p, t) = \sum_{i=1}^n E_i(\sigma, \varepsilon, v_p) \frac{d\varepsilon_i}{dt} dt + \sum_{i=1}^n \left\{ h_i[\sigma(t), \varepsilon(t), v_p(t), t] e^{\frac{-\Delta t}{\tau_i}} + \frac{E_i(\sigma, \varepsilon, v_p) \left(1 - e^{\frac{-\Delta t}{\tau_i}}\right)}{\left(\frac{\Delta t}{\tau_i}\right)} \frac{d\varepsilon_i}{dt} dt \right\} \quad (1)$$

$$E_i(\sigma, \varepsilon, v_p) = a \left(1 + b \left(\frac{\sigma}{\sigma_{max}} \right)^c \right) \quad (2)$$

$$\tau_i(\eta_i, E_i, t) = \frac{10\eta_i(\sigma, \varepsilon, v_p)}{E_i(\sigma, \varepsilon, v_p)} \quad (4)$$

$$h_i[\sigma(t), t] e^{\frac{-\Delta t}{\tau}} = \int_0^t \left\{ E_i(\sigma) e^{\frac{-(t-\Delta t+\tau)}{\tau_i}} \right\} \frac{d\varepsilon_i}{dt} dt \quad (3)$$

$$\eta_i(\sigma, t) = d \left(1 + e \left(\frac{\sigma}{\sigma_{max}} \right)^f \right) \quad (5)$$

Where σ is stress, ε is strain, v_p is wave velocity, E_i is the relaxation modulus, η_i is viscosity, t is time, τ is the relaxation time, and $a - f$ are parameters. We also used similar equations for E_i and η_i as a function of ε and v_p instead of σ .

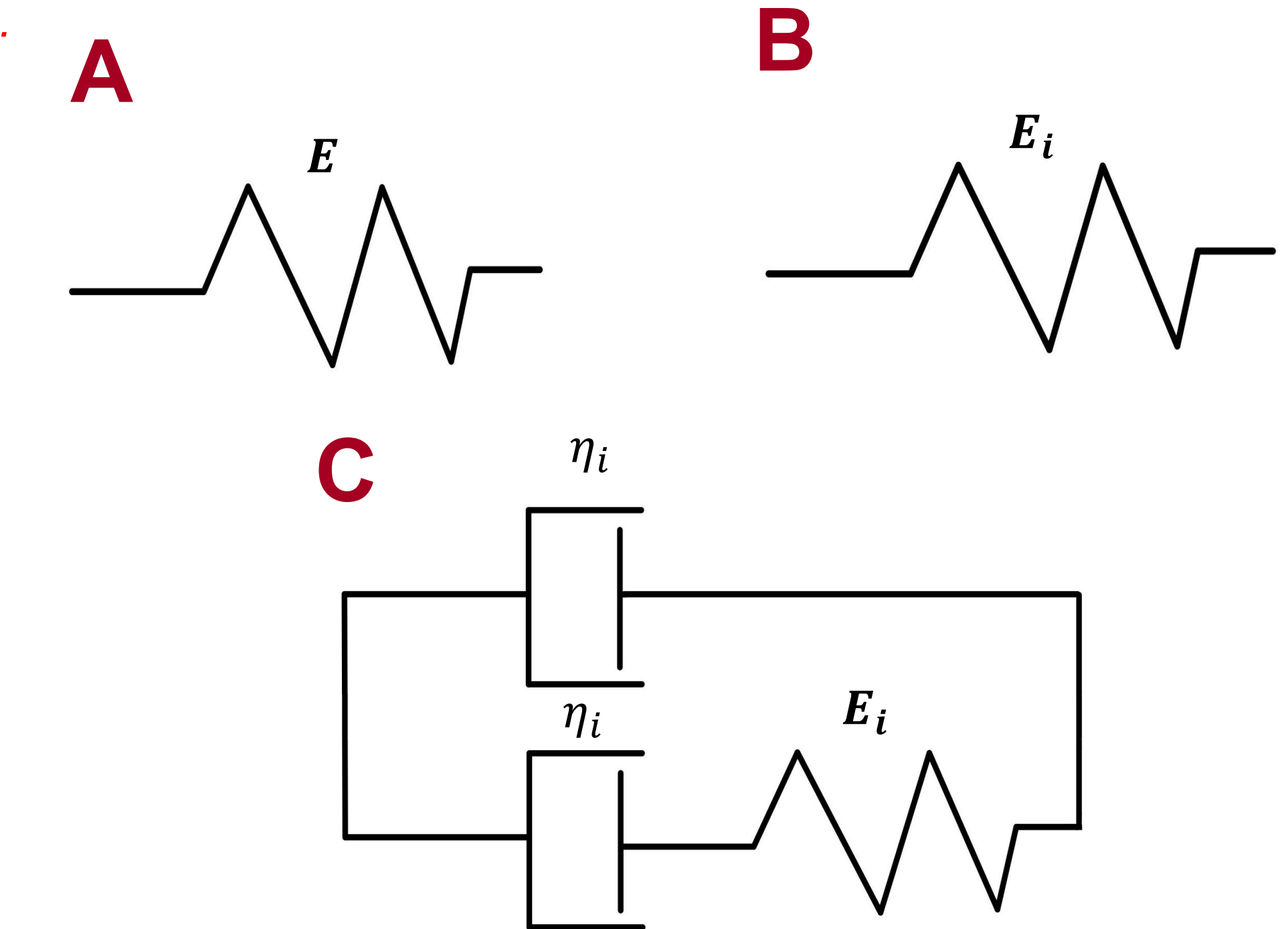


Figure 8: Model illustrations. (A) linear elastic. (B) non-linear elastic. (C) non-linear viscoelastic with changing viscosity

Modeling Results

- The simple rheological models (elastic and non-linear elastic) were unable to accurately represent the mechanical behavior of rocks. Even the more complex models, such as the non-linear viscoelastic model with changing viscosity, struggled to fully capture the observed data (**Fig. 9**).

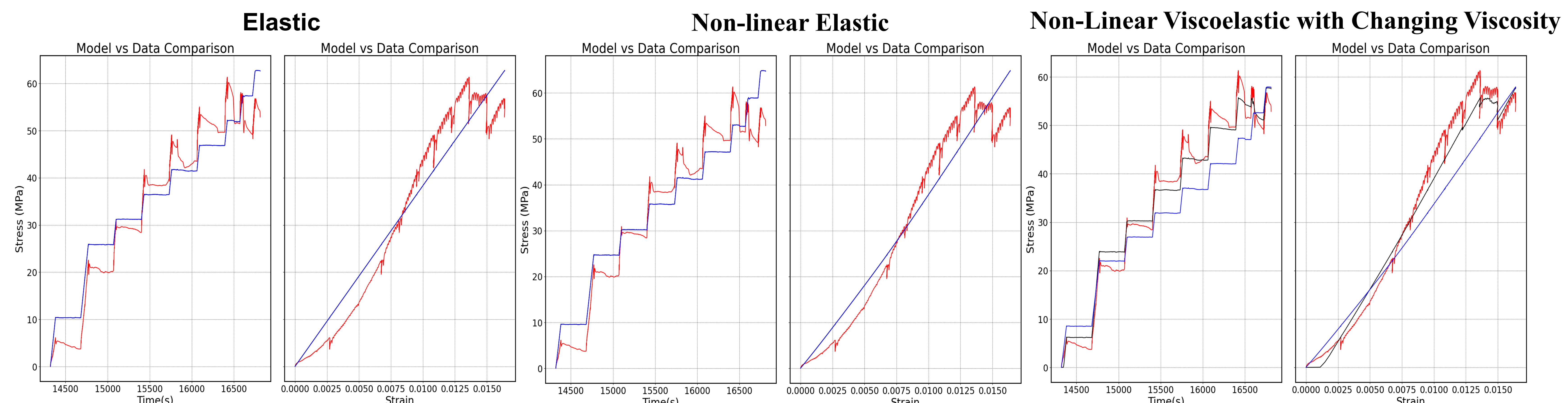


Figure 9: Modeling results for the elastic, non-linear elastic, and non-linear viscoelastic models.

Conclusion

Research Questions

- Rocks with different physical properties exhibit different mechanical properties.
- Simple rheological models fail to adequately capture these complexities. Even more advanced models struggle to align with the observed data.

Future Work

- A damage variable derived from the acoustic data is needed for capturing micro-crack propagation and quantifying the degradation of mechanical properties.
- Applying UMAP, a dimensionality reduction technique, to the waveform data to define new E_i and η_i could lead to a more accurate representation of the evolving mechanical behavior (**Fig. 10 & 11**).

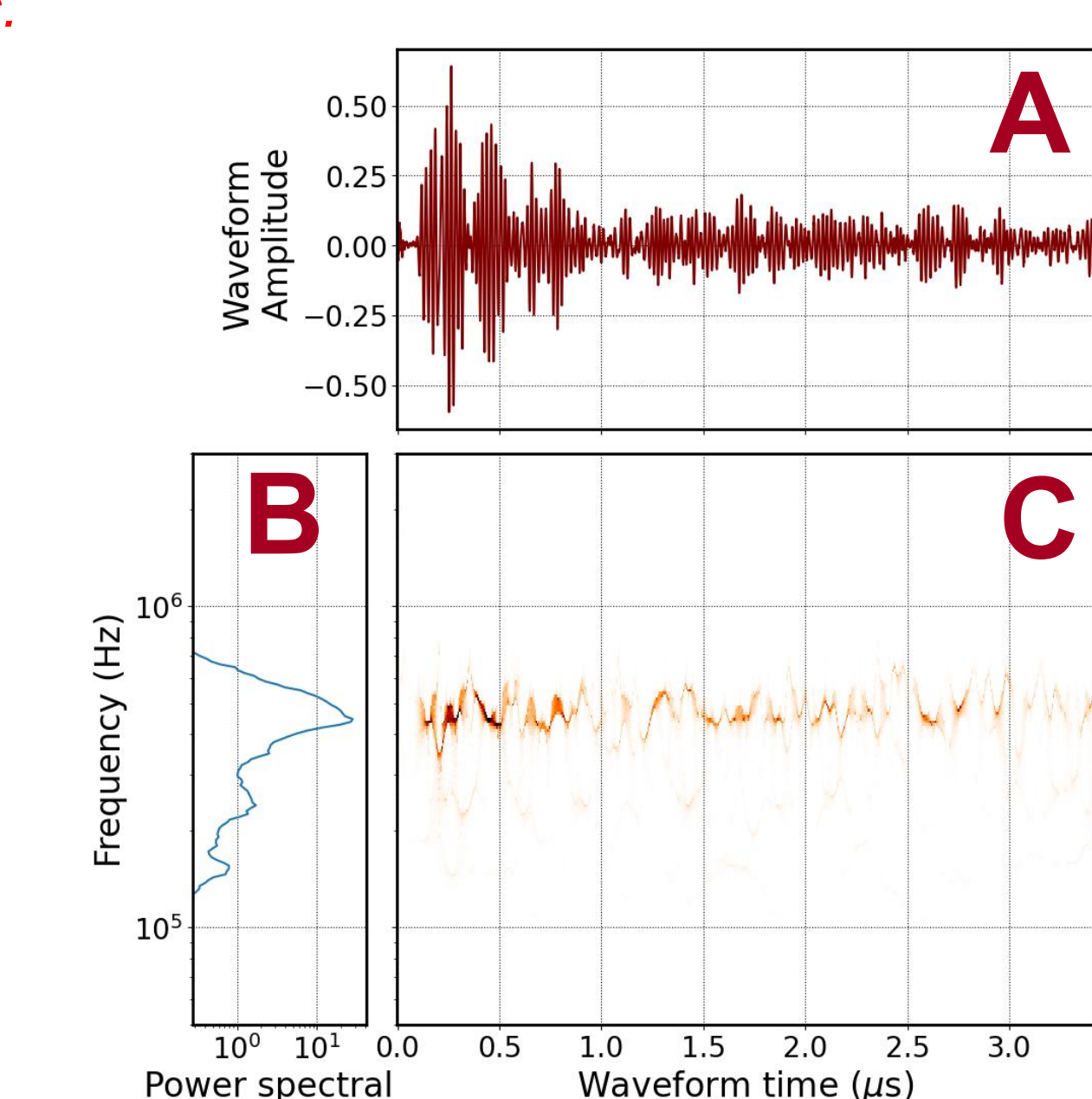


Figure 10: (A) Single waveform, (B) shows how the power of the signal across different frequencies, and (C) shows how the power of the signal evolves over time.

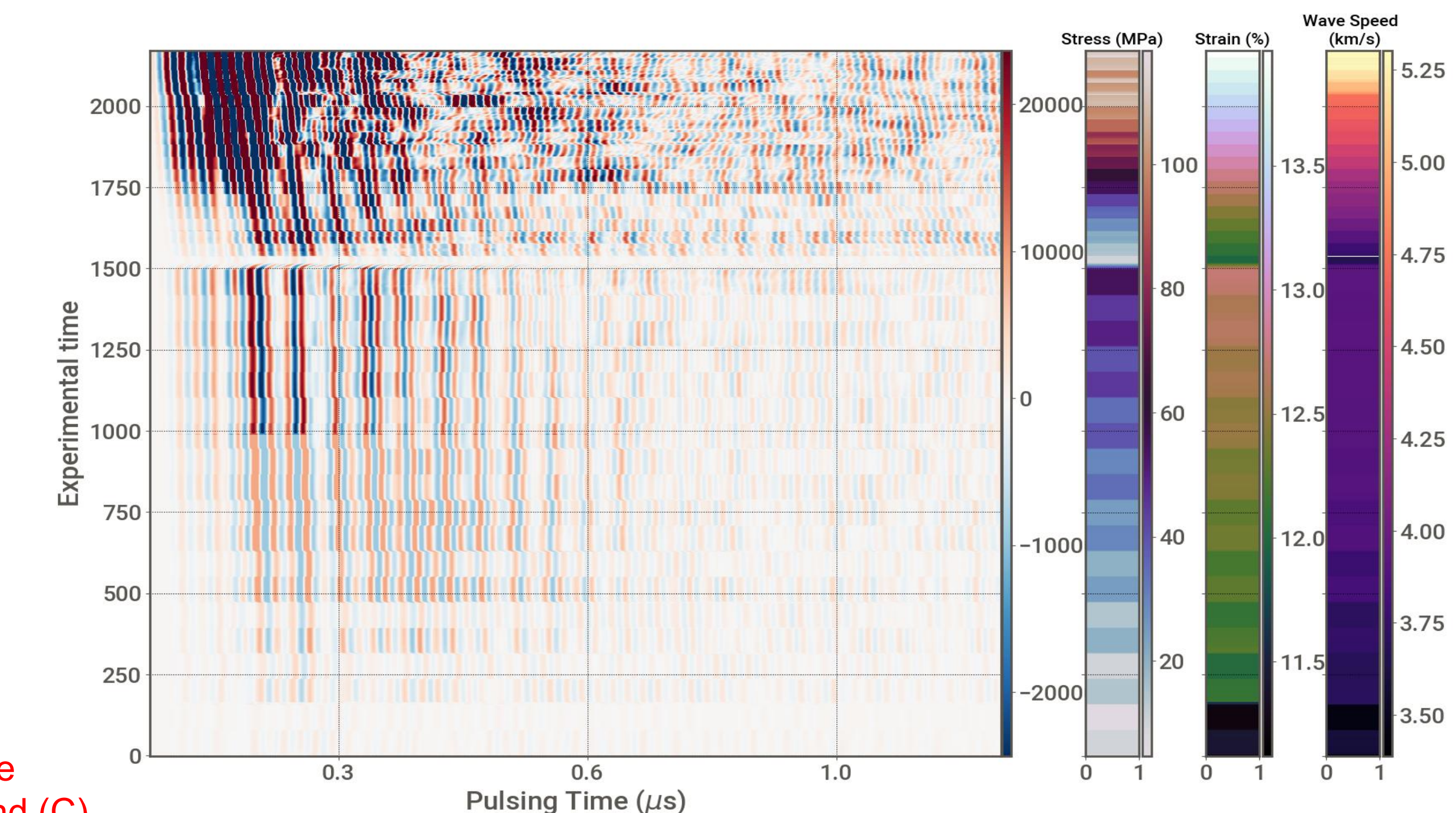


Figure 11: Stacked waveforms for experiment p5938.

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